

An implicit-weight mixture-effect SER: clean derivation

MixSER notes

June 12, 2026

This note derives, from scratch, the mixture-effect single-effect regression (SER) used in **MixSER**, with the mixing weights learned *implicitly* by empirical Bayes – the device behind **ashr** [1], solved by the convex optimiser **mixsqp** [2] – rather than through a Dirichlet prior. There are two layers: a *coding-model* mixture at the SNP level (§3, identifiable, the recommended core) and a *per-individual* mixture within a SNP (§4). Both reduce to the same convex program (§2). Throughout, σ^2 is the residual variance and effects are in residual-SD units.

1 The single-effect building block

Let $x \in \mathbb{R}^n$ be one centred (and scaled) predictor and $y \in \mathbb{R}^n$ a centred response. The Bayesian simple regression

$$y \mid \beta \sim \mathcal{N}(x\beta, \sigma^2 I_n), \quad \beta \sim \mathcal{N}(0, \sigma_0^2) \quad (1)$$

has a closed-form marginal. With $\hat{\beta} = x^\top y / x^\top x$ and $s^2 = \sigma^2 / x^\top x$, completing the square in β gives $p(y \mid \sigma_0^2) = \mathcal{N}(y; 0, \sigma^2 I + \sigma_0^2 x x^\top)$, and the log Bayes factor against $\sigma_0^2 = 0$ is the Wang–Sarkar–Carbonetto–Stephens form [3],

$$\log \text{BF}(\sigma_0^2) = \frac{1}{2} \log \frac{s^2}{\sigma_0^2 + s^2} + \frac{1}{2} \hat{\beta}^2 \left(\frac{1}{s^2} - \frac{1}{\sigma_0^2 + s^2} \right), \quad (2)$$

with Gaussian posterior

$$\beta \mid y \sim \mathcal{N}(\mu, \tau^2), \quad \tau^2 = (\sigma_0^{-2} + s^{-2})^{-1}, \quad \mu = \frac{\tau^2}{s^2} \hat{\beta}. \quad (3)$$

The single-effect regression [3] places one effect among p candidates with prior weights γ_j (uniform $1/p$): posterior inclusion $\alpha_j \propto \gamma_j \text{BF}_j$, SER log Bayes factor $\log \sum_j \gamma_j \text{BF}_j$, and prior variance by empirical Bayes $\hat{\sigma}_0^2 = \arg \max_{\sigma_0^2} \log \sum_j \gamma_j \text{BF}_j(\sigma_0^2)$.

2 Empirical-Bayes mixture weights (the ash/mixsqp core)

Suppose we have T “observations” t and K components, with likelihood $L_{tk} \geq 0$ of observation t under component k , and weights $w \in \Delta^{K-1} = \{w \geq 0, \mathbf{1}^\top w = 1\}$. The marginal log-likelihood

$$\ell(w) = \sum_{t=1}^T \log \left(\sum_{k=1}^K w_k L_{tk} \right) \quad (4)$$

is maximised over the simplex.

Proposition 1 (Convexity). $\ell(w)$ is concave on Δ , so $\hat{w} = \arg \max_{w \in \Delta} \ell(w)$ is a convex program.

Proof. Each $w \mapsto L_t w$ is linear; $\log$ is concave and nondecreasing; the composition is concave, a sum of concave functions is concave, and Δ is convex. $\square$

This is the maximum-likelihood mixture-proportions problem of `ash` [1], solved by SQP in `mixsqp` [2]. Two facts we reuse:

Lemma 1 (Row scaling is irrelevant). *Replacing $L_{tk} \leftarrow c_t L_{tk}$ ($c_t > 0$) leaves $\arg \max \ell$ unchanged (it shifts ℓ by $\sum_t \log c_t$). Hence we may feed Bayes factors.*

Lemma 2 (EM is an MM algorithm for (4)). *With $r_{tk} = w_k L_{tk} / \sum_{k'} w_{k'} L_{tk'}$, the update $w_k \leftarrow \frac{1}{T} \sum_t r_{tk}$ increases ℓ monotonically to $\hat{w}$.*

Proof. By Jensen $\ell(w') - \ell(w) \geq \sum_{t,k} r_{tk} \log(w'_k / w_k)$, maximised over Δ at $w'_k = \frac{1}{T} \sum_t r_{tk}$; the MM principle gives monotonicity and convergence to the convex optimum. $\square$

The implementation uses `mixsqp` when available and Lemma 2 as a fallback. The responsibility r_{tk} is the posterior weight that observation t belongs to component k .

3 Level 1: the coding-model mixture SER

A genotype $g \in \{0, 1, 2\}$ gives codings $x^A(g) = g$, $x^D(g) = \mathbf{1}\{g \geq 1\}$, $x^R(g) = \mathbf{1}\{g = 2\}$. Inside one IBSS single effect we hold a centred partial residual y and, for variant j and coding m , the centred-scaled x_j^m .

Per-coding hyperparameters (as in SuSiE).

$$\hat{\sigma}_{0m}^2 = \arg \max_V \log \sum_j \gamma_j \text{BF}_{jm}(V), \quad \text{BF}_{jm}(V) = \exp((2) \text{ for } x_j^m), \quad (5)$$

allowing $\hat{\sigma}_{0m}^2 = 0$ (coding off) when that does not decrease the objective – this is what makes the procedure additive-leaning without tuning.

Implicit model weights (this replaces the Dirichlet). Augment with an explicit null ($\text{BF}_{j0} \equiv 1$) of fixed prior mass $w_\emptyset$ (default 0, as `susier`'s `null_weight`). The coding weights $w = (w_A, w_D, w_R) \in \Delta^2$ are learned by §2 with the p variants as observations and codings as components,

$$\hat{w} = \arg \max_{w \in \Delta^2} \sum_{j=1}^p \log \left(\sum_{m \in \{A, D, R\}} w_m \text{BF}_{jm}(\hat{\sigma}_{0m}^2) \right), \quad (6)$$

convex by Prop. 1, valid with Bayes factors by Lemma 1. It is self-regularising: a null variant has $\text{BF}_{jm} \approx 1$, contributing $\log(\sum_m w_m) = 0$, so only signal variants move $\hat{w}$. (Genome-wide: solve (6) once over many SNPs and freeze $\hat{w}$ – the pooled mode.)

Posteriors. By Bayes' rule, for $m \in \{A, D, R\}$,

$$\alpha_{jm} = \frac{\gamma_j(1 - w_\emptyset)\hat{w}_m \text{BF}_{jm}}{w_\emptyset + \sum_{j'm'} \gamma_{j'}(1 - w_\emptyset)\hat{w}_{m'} \text{BF}_{j'm'}}, \quad (7)$$

giving PIP $\alpha_j = \sum_m \alpha_{jm}$, coding posterior $\Pr(m | j) = \alpha_{jm}/\alpha_j$, SER log-BF $\log \sum_j \gamma_j \sum_m \hat{w}_m \text{BF}_{jm}$, and fitted effect $\hat{y} = \sum_{j,m} \alpha_{jm} x_j^m \mu_{jm}$ – exactly the inputs IBSS consumes.

Proposition 2 (Exact reduction to SuSiE). *With one coding and $w_\emptyset = 0$, (7) is $\alpha_j = \gamma_j \text{BF}_j / \sum_{j'} \gamma_{j'} \text{BF}_{j'}$ with $\hat{\sigma}_0^2$ from (5): the standard additive SuSiE SER, identically (verified: $\max_j |\alpha_j^{\text{mix}} - \alpha_j^{\text{SuSiE}}| = 0$).*

Level 1 is a statement about the conditional means $\mathbb{E}[y | g]$, identifiable at $O(\theta^2)$; it is where every coding decision should be made.

4 Level 2: per-individual mixture, without a Dirichlet

At a single variant, each individual i carries a latent coding $z_i \in \{1, \dots, M\}$ with fractions $\pi \in \Delta^{M-1}$:

$$z_i \sim \text{Cat}(\pi), \quad y_i | z_i=m \sim \mathcal{N}(x_i^m \beta_m, \sigma^2), \quad \beta_m \sim \mathcal{N}(0, \sigma_{0m}^2). \quad (8)$$

Treat $\pi, \sigma_{0m}^2, \sigma^2$ as hyperparameters estimated by empirical Bayes – so there is **no** $q(\pi)$ and **no** Dirichlet KL. With the mean-field posterior $\prod_i q(z_i) \prod_m q(\beta_m)$, $q(z_i=m) = r_{im}$, $q(\beta_m) = \mathcal{N}(\mu_m, \tau_m^2)$, the evidence lower bound is

$$\begin{aligned} \mathcal{F} = & -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i y_i^2 + \sum_i \sum_m r_{im} \left[\log \pi_m - \frac{1}{2\sigma^2} (-2x_i^m y_i \mu_m + (x_i^m)^2 (\mu_m^2 + \tau_m^2)) \right] \\ & - \sum_i \sum_m r_{im} \log r_{im} - \sum_m \text{KL}(\mathcal{N}(\mu_m, \tau_m^2) \| \mathcal{N}(0, \sigma_{0m}^2)), \end{aligned} \quad (9)$$

with $\text{KL} = \frac{1}{2}(\log \frac{\sigma_{0m}^2}{\tau_m^2} + \frac{\tau_m^2 + \mu_m^2}{\sigma_{0m}^2} - 1)$. The two data constants are kept so $\mathcal{F}$ is a genuine lower bound on $\log p(y)$.

Coordinate ascent.

$$q(\beta_m) : \tau_m^2 = \left(\sigma_{0m}^{-2} + \sigma^{-2} \sum_i r_{im} (x_i^m)^2 \right)^{-1}, \quad \mu_m = \frac{\tau_m^2}{\sigma^2} \sum_i r_{im} x_i^m y_i, \quad (10)$$

$$q(z_i) : \log r_{im} \propto \log \pi_m - \frac{1}{2\sigma^2} (-2x_i^m y_i \mu_m + (x_i^m)^2 (\mu_m^2 + \tau_m^2)), \quad (11)$$

$$\sigma_{0m}^2 : \sigma_{0m}^2 = \mu_m^2 + \tau_m^2, \quad (12)$$

$$\pi : \pi = \arg \max_{\pi \in \Delta} \sum_i \log (\sum_m \pi_m L_{im}), \quad L_{im} = \exp(\mathbb{E}_q \log \mathcal{N}(y_i; x_i^m \beta_m, \sigma^2)), \quad (13)$$

$$\sigma^2 : \sigma^2 = \frac{1}{n} \sum_i \sum_m r_{im} (y_i^2 - 2x_i^m y_i \mu_m + (x_i^m)^2 (\mu_m^2 + \tau_m^2)). \quad (14)$$

Equation (11) is the softmax of (9); (12) the EB maximiser of the KL term; (10) the weighted (3); and (14) the stationary point $\partial \mathcal{F} / \partial \sigma^2 = 0$. **Estimating σ^2 is essential for a standalone single-SNP fit:** with σ^2 fixed too large, the responsibilities collapse to the prior π (everyone looks like a blend); EB recovers the noise level and with it sharp responsibilities. (Inside SuSiE, σ^2 is supplied and (14) is skipped.)

Why the π -update *is* the mixsqp program. The π -terms of (9) are $\sum_{i,m} r_{im} \log \pi_m$; at optimum $r_{im} \propto \pi_m L_{im}$ (11), so profiling r out turns them into $\sum_i \log \sum_m \pi_m L_{im}$ – exactly (4) with individuals as observations and codings as components. This is the Dirichlet’s replacement: where the old code formed $\mathbb{E}[\log \pi_m]$ from a Dirichlet posterior, we set $\log \pi_m$ from the mixsqp solution; the EM form is $\pi_m = \frac{1}{n} \sum_i r_{im}$.

Initialisation. The uniform point $r_{im} \equiv 1/M$ is a saddle of (9), and random starts can fall into degenerate optima. Initialise from the *pure single-coding* least-squares effects $\mu_m^{(0)} = \sum_i x_i^m y_i / \sum_i (x_i^m)^2$ and the responsibilities they imply: this breaks symmetry in a data-driven direction and attains the highest ELBO (a few random restarts are kept as insurance, selecting the best $\mathcal{F}$).

Reading r . r_{im} is sharp (near 0/1) only for individuals whose codings genuinely differ – at $g = 2$, where additive and recessive predictions diverge. At $g = 0$ all codings predict the baseline, the data are uninformative, and r_i *correctly* returns the prior π . Summarise r stratified by genotype, never pooled over g .

5 Why decisions live at Level 1, and the pooling requirement

The per-individual fractions are weakly identified: with $g \in \{0, 1, 2\}$ the data are summarised by $p(y | g)$, and the mixture’s signal over a single-coding mean is an excess within-genotype variance $\text{Var}(y | g=1) = \sigma^2 + \pi(1 - \pi)\theta^2 - O(\theta^2)$ in a second moment, $O(\theta^4)$ in information, confounded with heteroscedasticity (full analysis in `identifiability_per_individual_mixture.md`). Hence:

1. **Selection is a Level-1 task.** The additive/dominant/recessive/mixture posterior is (7), each model scored by a proper marginal likelihood. One must not decide pure-vs-mixture by a per-locus point estimate of π : (8) nests the pure models (put π on a vertex), so $\max_{\pi} p(y | \pi) \geq p_{\text{pure}}(y)$ always – the maximised- π mixture wins by construction (the VEB-pi pathology; integrating π under a Dirichlet was the old fix).
2. **π must be pooled across loci.** Within one locus even the convex MLE (13) places interior mass to fit within-genotype noise. The honest EB cure – the one that makes ash valid – estimates *one* π across many loci by stacking the L_{im} of (13) into a single (4); a pure-additive locus then contributes under the shared π and cannot manufacture a mixture. Per-individual r_{im} are reported conditionally on this pooled $\hat{\pi}$; per-locus π is exploratory only.

The Dirichlet’s two roles – an Occam factor and regularising π – are taken over by Level-1’s pooled marginal likelihood and by cross-locus EB pooling of (13), with no prior to tune.

Defaults and what ships

Codings $\{A, D, R\}$ (+ optional null, $w_{\emptyset} = 0$); σ_{0m}^2 by (5); coding weights $\hat{w}$ by (6) (`per_ser` or pooled); π by (13); σ^2 by (14); solver `mixsqp` with EM fallback. Implemented in `ser_mix_eb()` (Level 1), `ser_pim_vb_eb()` (Level 2, pure-coding init, EB σ^2), and `estimate_mixture_weights()`.

References

- [1] M. Stephens (2017). False discovery rates: a new deal. *Biostatistics* 18(2):275–294.

- [2] Y. Kim, P. Carbonetto, M. Stephens, M. Anitescu (2020). A fast algorithm for maximum likelihood estimation of mixture proportions using sequential quadratic programming. *JCGS* 29(2):261–273.
- [3] G. Wang, A. Sarkar, P. Carbonetto, M. Stephens (2020). A simple new approach to variable selection in regression, with application to genetic fine mapping. *JRSS-B* 82(5):1273–1300.